AI SOFTWARE DEVELOPMENT INTERNSHIP

EMAN MASOOD (BSSE)

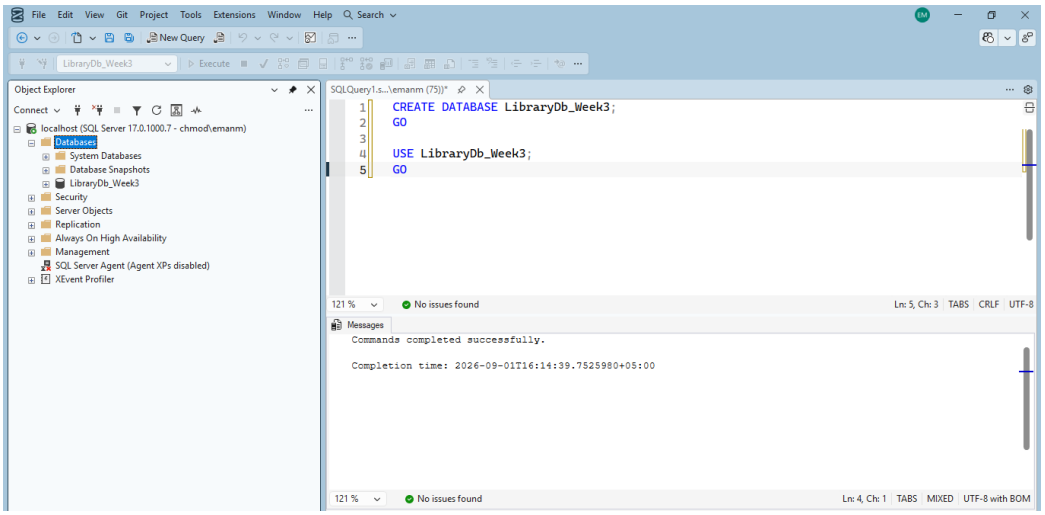
Week 3: SQL Server & Entity Framework Core • API/Angular Integration Git & GitHub Workflow • AI & Python Foundations

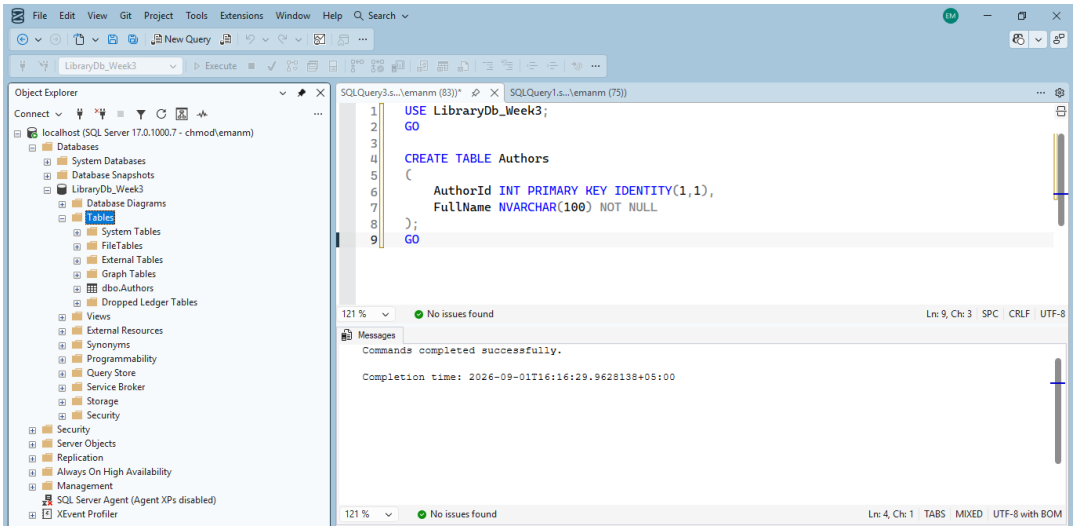
PART A — RELATIONAL DATABASE DESIGN (BEYOND BASIC TABLES)

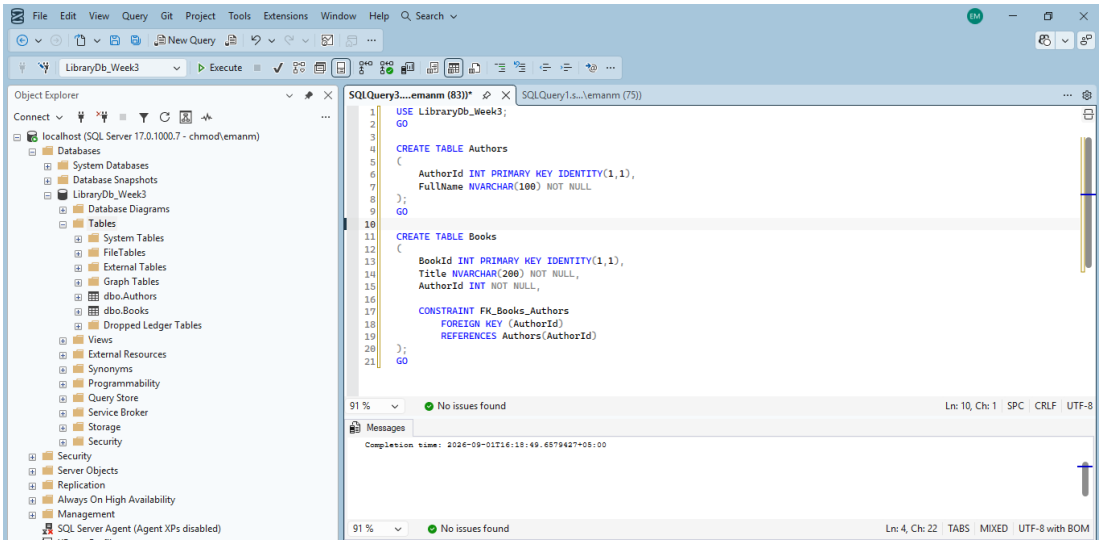
Practice Tasks

1. **Create an Authors table (AuthorId PK, FullName) and a Books table (BookId PK, Title, AuthorId FK referencing Authors).**

Output:

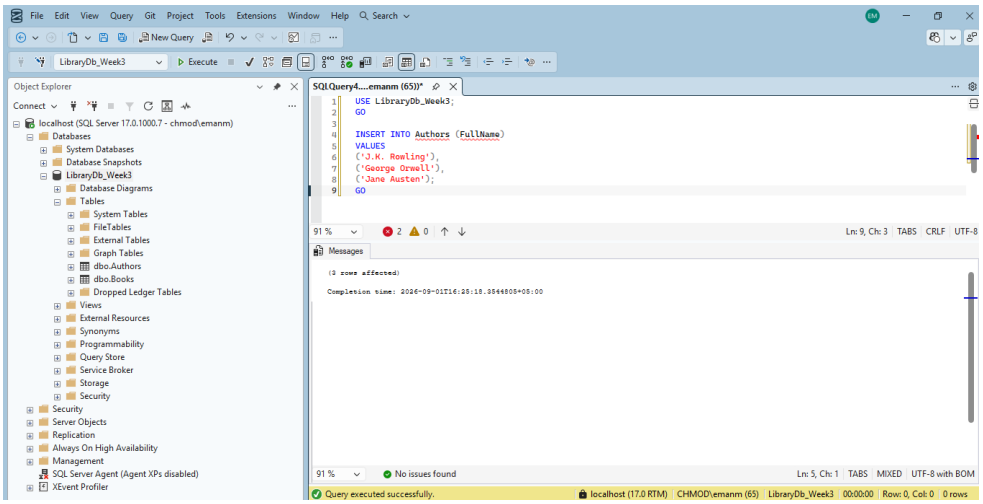


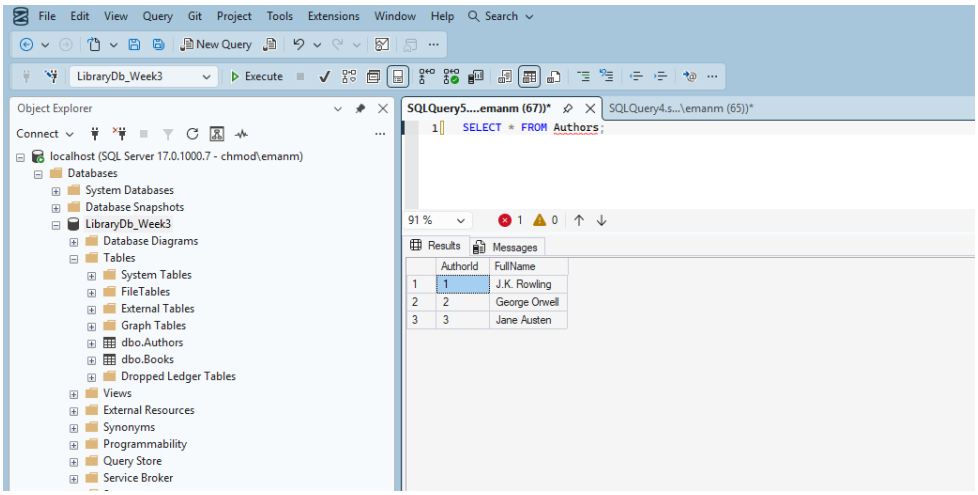


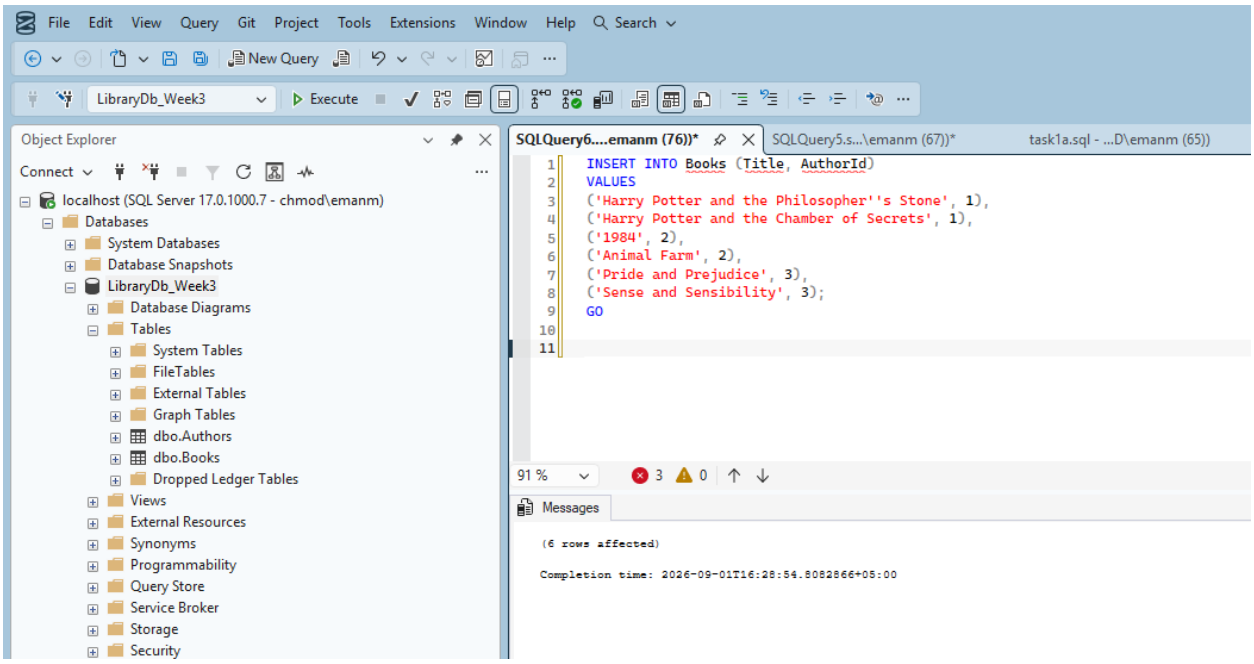


2. **Insert 3 authors and 6 books split between them**

Output:




File Edit View Query Git Project Tools Extensions Window Help Search
New Query
LibraryDb_Week3
Execute
Object Explorer
Connect
localhost (SQL Server 17.0.1000.7 - chmod\emanm)
Databases
System Databases
Database Snapshots
LibraryDb_Week3
Database Diagrams
Tables
System Tables
FileTables
External Tables
Graph Tables
dbo.Authors
dbo.Books
Dropped Ledger Tables
Views
External Resources
Synonyms
Programmability
Query Store
Service Broker
SQLQuery5....emanm (67))*
SQLQuery4.s...\emanm (65))*
1 SELECT * FROM Authors;
91 %
Results
Messages
AuthorId FullName
1 1 J.K. Rowling
2 2 George Orwell
3 3 Jane Austen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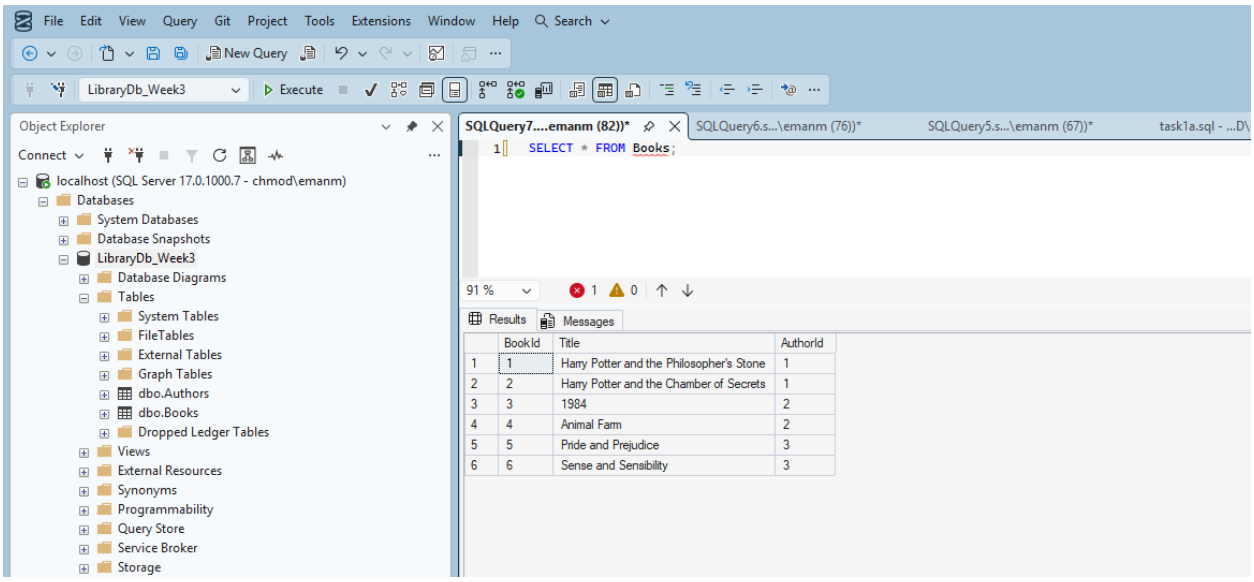
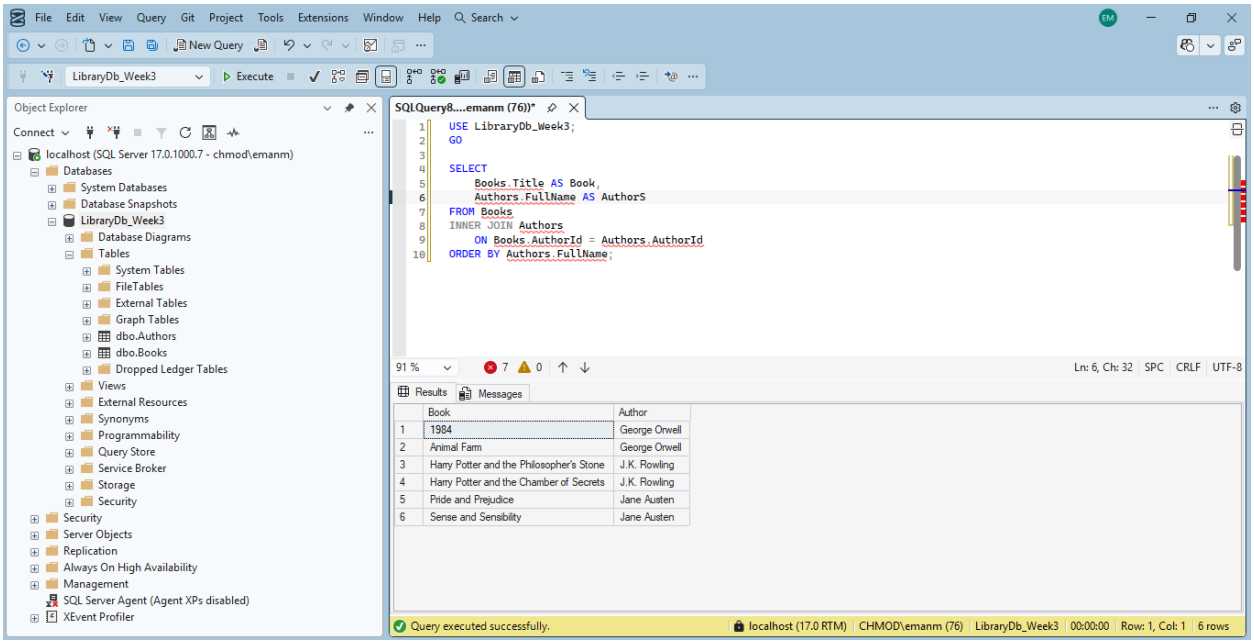
Views
External Resources
Synonyms
Programmability
Query Store
Service Broker
Storage
Security
SQLQuery6....emanm (76))*
SQLQuery5.s...\emanm (67))*
task1a.sql - ...D\emanm (65))
INSERT INTO Books (Title, AuthorId)
VALUES
('Harry Potter and the Philosopher''s Stone', 1),
('Harry Potter and the Chamber of Secrets', 1),
('1984', 2),
('Animal Farm', 2),
('Pride and Prejudice', 3),
('Sense and Sensibility', 3);
GO
91 %
3
0
Messages
(6 rows affected)
Completion time: 2026-09-01T16:28:54.8082866+05:00



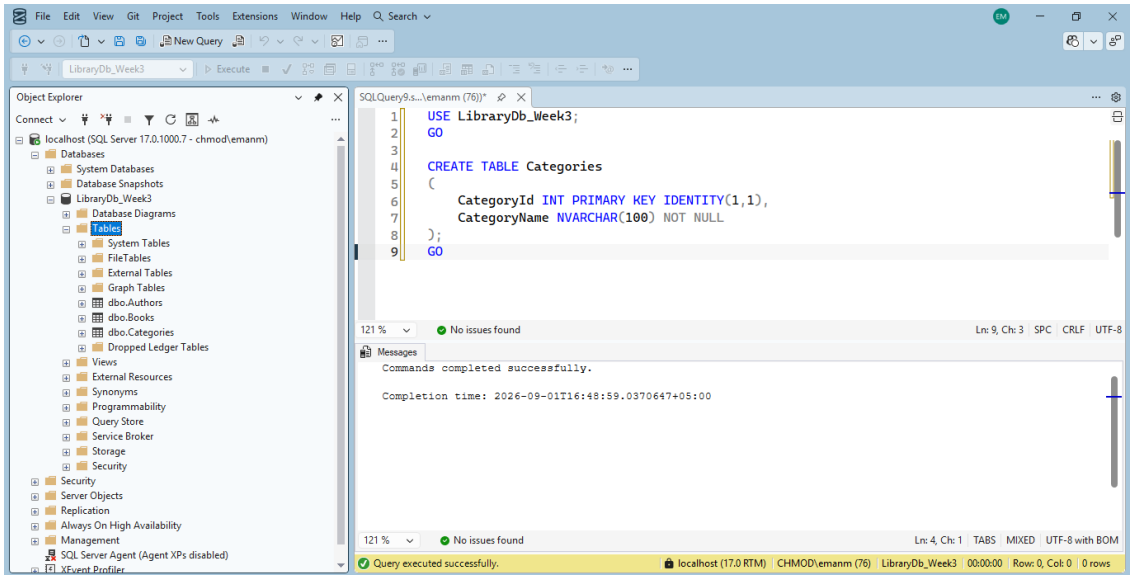
3. **Write a JOIN query that lists every book together with its author's name in one result set.**

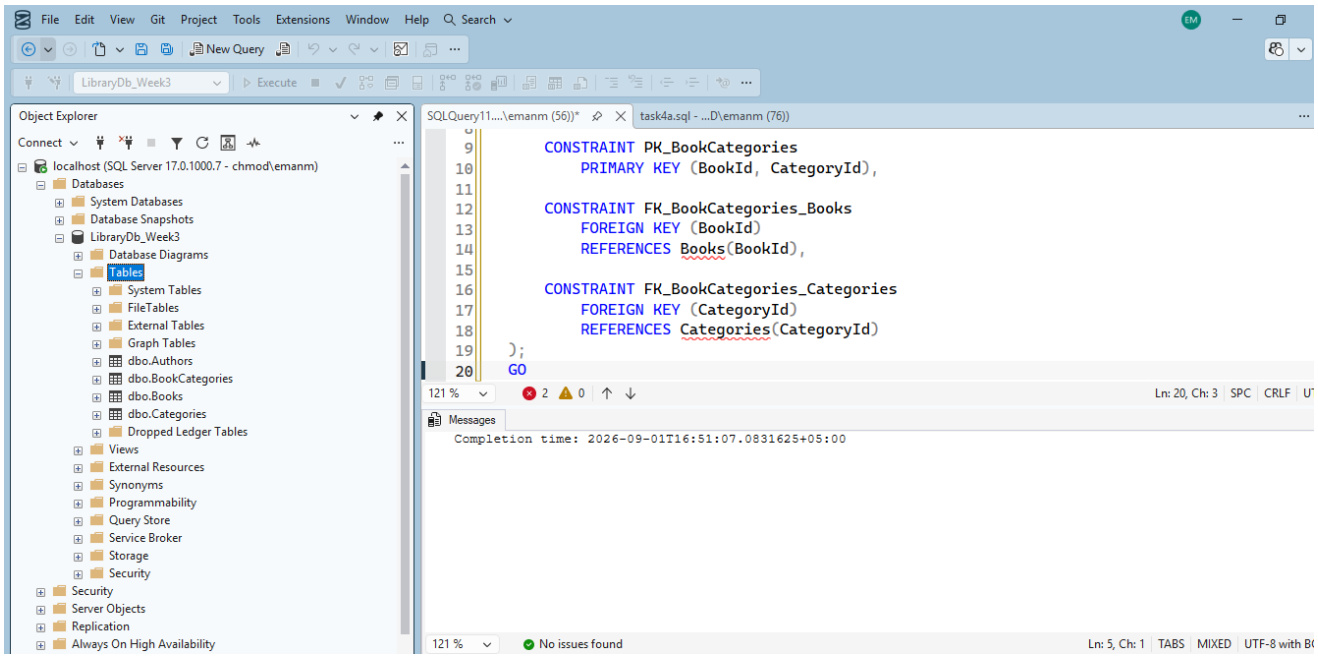
Output:

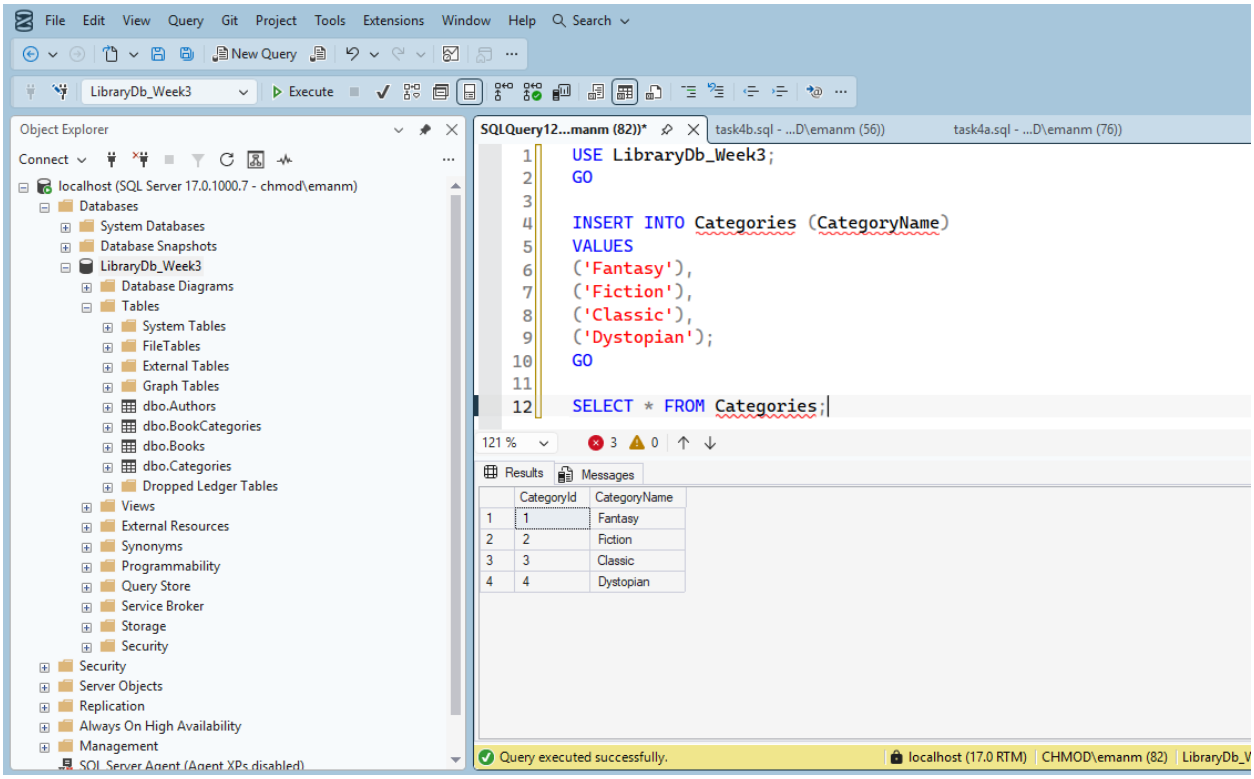


4. **Create a Categories table and a link table BookCategories (BookId, CategoryId) to represent many-to-many, then insert sample rows connecting books to more than one category.**

Output:



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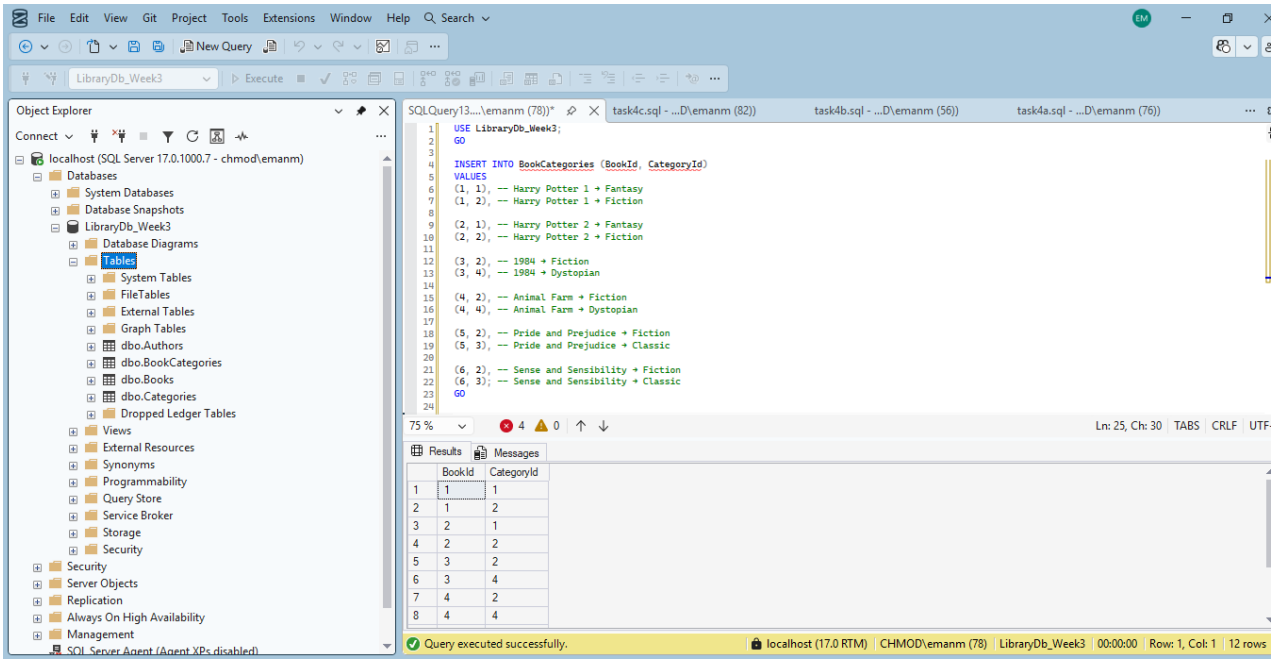
dbo.BookCategories
dbo.Books
dbo.Categories
Dropped Ledger Tables
Views
External Resources
Synonyms
Programmability
Query Store
Service Broker
Storage
Security
Security
Server Objects
Replication
Always On High Availability
Management
SQL Server Agent (Agent XPs disabled)
SQLQuery12...manm (82))*
task4b.sql - ...D\emanm (56))
task4a.sql - ...D\emanm (76))
USE LibraryDb_Week3;
GO
INSERT INTO Categories (CategoryName)
VALUES
('Fantasy'),
('Fiction'),
('Classic'),
('Dystopian');
GO
SELECT * FROM Categories;
121 %
Results
Messages
CategoryId CategoryName
1 1 Fantasy
2 2 Fiction
3 3 Classic
4 4 Dystopian
Query executed successfully.
localhost (17.0 RTM)
CHMOD\emanm (82)
LibraryDb_W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Storage
Security
Security
Server Objects
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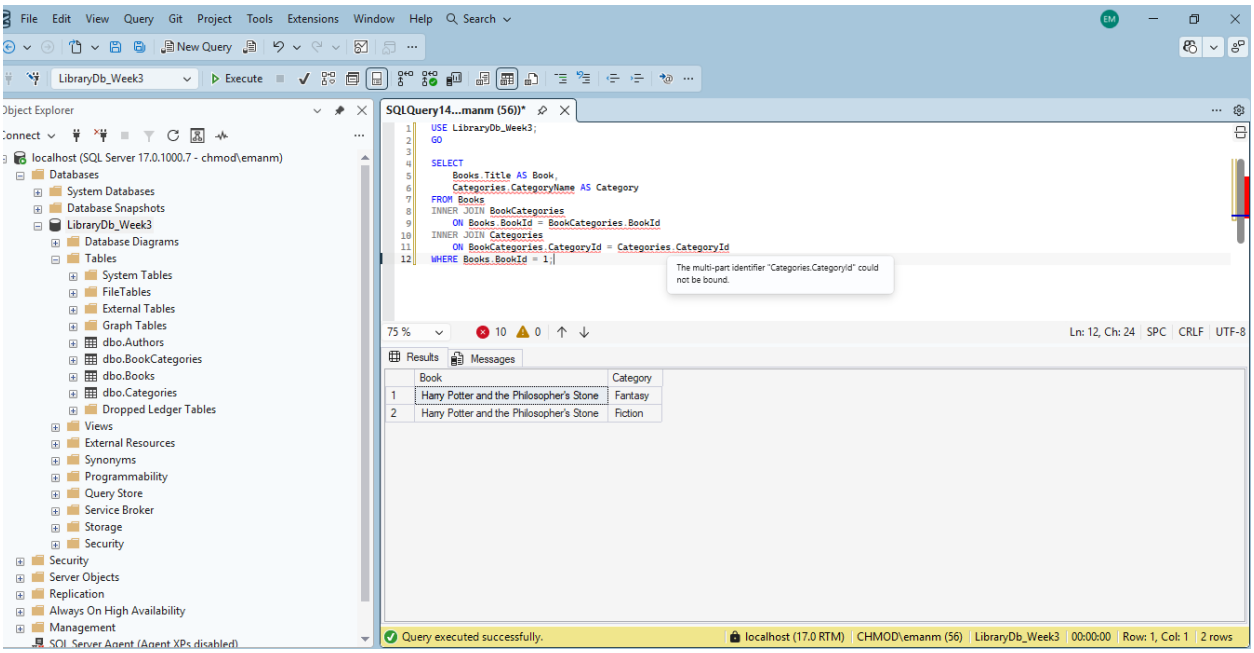
Always On High Availability
Management
SQL Server Agent (Agent XPs disabled)
SQLQuery13....\emanm (78))*
task4c.sql - ...D\emanm (82))
task4b.sql - ...D\emanm (56))
task4a.sql - ...D\emanm (76))
USE LibraryDb_Week3;
GO
INSERT INTO BookCategories (BookId, CategoryId)
VALUES
(1, 1), -- Harry Potter 1 → Fantasy
(1, 2), -- Harry Potter 1 → Fiction
(2, 1), -- Harry Potter 2 → Fantasy
(2, 2), -- Harry Potter 2 → Fiction
(3, 2), -- 1984 → Fiction
(3, 4), -- 1984 → Dystopian
(4, 2), -- Animal Farm → Fiction
(4, 4), -- Animal Farm → Dystopian
(5, 2), -- Pride and Prejudice → Fiction
(5, 3), -- Pride and Prejudice → Classic
(6, 2), -- Sense and Sensibility → Fiction
(6, 3); -- Sense and Sensibility → Classic
GO
75 %
Ln: 25, Ch: 30
TABS
CRLF
Results
Messages
BookId CategoryId
1 1 1
2 1 2
3 2 1
4 2 2
5 3 2
6 3 4
7 4 2
8 4 4
Query executed successfully.
localhost (17.0 RTM)
CHMOD\emanm (78)
LibraryDb_Week3
00:00:00
Row: 1, Col: 1
12 rows

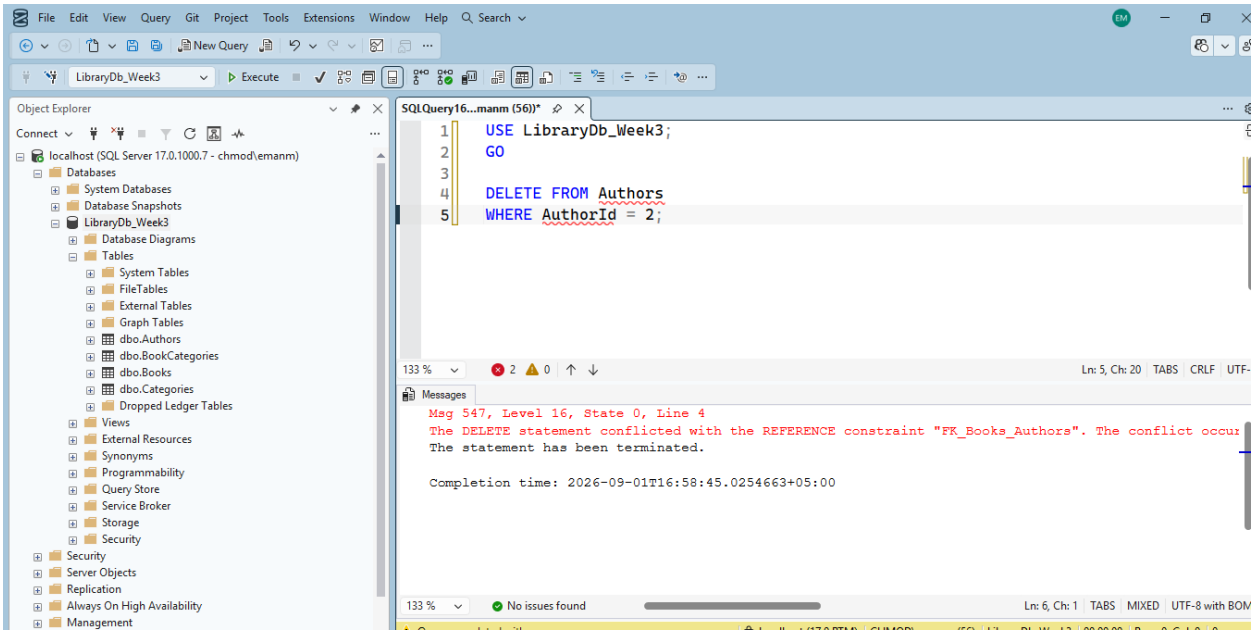
5. **Write a query that returns all categories for one specific book using the link table.**

Output:



6. **Try deleting an Author that still has Books linked to it and read the error SQL Server gives you — explain in your own words why it refused.**

Output:



SQL Server refused to delete the author because the author is still referenced by records in the Books table. The Books table has a foreign key (AuthorId) that references the Authors table. Deleting the author would leave the related books without a valid author, so SQL Server prevented the deletion to maintain referential integrity.